Level A: Easy Questions (2 Marks Each)

Q1. What is a process in computing?

A process is a program in execution, including its code, data, and resources required for execution.

Q2. Define the principle of concurrency.

Concurrency allows multiple processes or threads to execute simultaneously, enhancing resource utilization and system efficiency.

Q3. What is the producer/consumer problem?

The producer/consumer problem involves two processes, where the producer generates data and the consumer uses it, requiring synchronization to avoid conflicts.

Q4. What is mutual exclusion in concurrency?

Mutual exclusion ensures that only one process accesses a shared resource at a time, preventing race conditions.

Q5. What is the critical section problem?

The critical section problem addresses the need to synchronize processes that access shared resources to ensure data consistency.

Q6. What is Peterson’s solution for process synchronization?

Peterson's solution is an algorithm ensuring mutual exclusion and process synchronization in a critical section using flags and a turn variable.

Q7. What is a semaphore in process synchronization?

A semaphore is a synchronization tool used to control access to shared resources through two atomic operations: wait() and signal().

Q8. Explain the test-and-set operation.

Test-and-set is an atomic instruction used for locking mechanisms, ensuring mutual exclusion by testing and modifying a lock variable.

Q9. What is the dining philosopher problem?

The dining philosopher problem illustrates challenges in resource allocation and deadlock prevention in concurrent processes sharing limited resources.

Q10. What is inter-process communication (IPC)?

IPC is a mechanism that allows processes to communicate and share data, either through message passing or shared memory.

Q11. Name two IPC schemes used in operating systems.

1. Message Passing: Processes communicate by sending messages.
2. Shared Memory: Processes share a common memory space for communication.

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Q12. What is process generation?

Process generation refers to the creation of new processes using system calls like fork() in Unix-based systems.

Q13. What is CPU scheduling?

CPU scheduling is the process of selecting which process gets CPU time to maximize efficiency and resource utilization.

Q14. Define the term 'scheduling criteria.'

Scheduling criteria are parameters used to evaluate scheduling algorithms, such as throughput, turnaround time, and waiting time.

Q15. What is preemptive scheduling?

Preemptive scheduling allows the CPU to be forcibly taken from a process for another process, ensuring responsiveness.

Q16. Give one example of a scheduling algorithm.

The Round Robin algorithm allocates fixed time slices (quantums) to processes in a cyclic order.

Q17. Define the term 'starvation' in scheduling.

Starvation occurs when a process waits indefinitely for resources due to lower priority.

Q18. What is a context switch?

A context switch is the process of saving and restoring the CPU state when switching between processes.

Q19. What is priority scheduling?

Priority scheduling assigns the CPU to the process with the highest priority, either preemptively or non-preemptively.

Q20. What is multiprocessor scheduling?

Multiprocessor scheduling involves assigning processes to multiple CPUs for parallel execution while minimizing contention and overhead.

21. Explain the process concept in an operating system. How is it different from a program?

Process

- A process is a running instance of a program, including program code, data, and execution state.
- It has dynamic states (Ready, Running, Waiting, Terminated) and is managed by the Process Control Block (PCB).

Process vs Program

- Process: Active, dynamic, and resource-consuming.
- Program: Static set of instructions stored on disk.

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22. Discuss the principle of concurrency and the challenges it introduces in process synchronization.

Concurrency

- Multiple processes/threads execute simultaneously, improving performance and resource utilization.

Challenges

- Race Conditions: Conflicts in shared resource access.
- Deadlocks: Processes stuck waiting indefinitely.
- Critical Section Problem: Ensuring safe access to shared data.

Solution

- Use semaphores, mutexes, and proper scheduling to synchronize processes.

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23. Describe the producer/consumer problem. How can it be solved using semaphores?

Producer/Consumer Problem

- Producers generate data into a shared buffer; consumers remove it.
- Challenge: Avoid buffer overflow or underflow.

Semaphore Solution

- Use semaphores for full slots, empty slots, and mutex to synchronize buffer access.

24. What is mutual exclusion, and why is it important in concurrent programming?

Definition

- Mutual exclusion ensures only one process accesses a shared resource at a time.

Importance

- Prevents data corruption, race conditions, and ensures consistency.

Implementation

- Techniques include mutexes, semaphores, and high-level constructs like monitors.

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25. Explain the critical section problem. What conditions must be satisfied to solve it?

Critical Section Problem

- Occurs when multiple processes access shared data simultaneously, risking inconsistency.

Solution Requirements

1. Mutual Exclusion: Only one process in the critical section.
2. Progress: Processes not in the critical section cannot block others.
3. Bounded Waiting: No process waits indefinitely.

Let me know if this shorter format works so I can continue with the rest!

No, continue

26. What are semaphores, and how are they used to control access to critical sections?

Semaphore

- A synchronization tool represented as an integer variable.
- Types: Binary semaphore (0/1) and Counting semaphore (integer).

Usage

Semaphore

- A synchronization tool represented as an integer variable.
- Types: Binary semaphore (0/1) and Counting semaphore (integer).

Usage

- Wait (P): Decrements the semaphore. If negative, process waits.
- Signal (V): Increments the semaphore, allowing waiting processes to proceed.
- Ensures mutual exclusion by controlling access to shared resources.

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27. Describe the dining philosopher problem. How does it illustrate the challenges of resource allocation in concurrent processes?

Dining Philosopher Problem

- Scenario: Philosophers sit at a table, alternating between eating and thinking, sharing limited forks (resources).
- Challenge: Avoid deadlock, starvation, and ensure proper resource allocation.

Illustration

- Represents problems of deadlock (all philosophers holding one fork) and starvation (low-priority philosopher never eats).
- Solutions include using semaphores, limiting simultaneous access, or introducing a waiter to manage resources.

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28. Explain the sleeping barber problem in the context of process synchronization. What are the potential deadlock issues?

Sleeping Barber Problem

- Scenario: A barber serves customers from a waiting room with limited chairs. If no customers, the barber sleeps.
- Challenge: Synchronizing customers entering, leaving, and the barber serving.

Deadlock and Solution

- Deadlock occurs if customers and barber wait indefinitely for each other.
- Solved using semaphores for available chairs and mutual exclusion for critical sections.

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29. Compare message passing and shared memory as models for inter-process communication (IPC).

Message Passing

- Processes exchange information via messages.
- Advantages: Simple, secure in distributed systems.
- Disadvantages: Slower due to message overhead.

Shared Memory

- Processes access a common memory space.
- Advantages: Fast and efficient.
- Disadvantages: Requires synchronization mechanisms to prevent race conditions.

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30. Discuss the evolution of scheduling algorithms in operating systems. How have modern algorithms improved system performance?

Evolution

- First-Come-First-Serve (FCFS): Simple but prone to long waiting times.
- Shortest Job Next (SJN): Reduces average wait but risks starvation.
- Round Robin (RR): Fair but high context switching overhead.

Modern Improvements

- Use of priority scheduling, multi-level queues, and dynamic adjustments improves responsiveness, throughput, and fairness.

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31. Explain the test-and-set instruction and how it is used to implement synchronization in concurrent processes. Discuss potential problems such as busy-waiting and propose alternative mechanisms to mitigate them.

Test-and-Set Instruction

- An atomic operation used to lock resources by testing and setting a variable in a single step.
- If the variable is 0, it sets it to 1 (locked). If 1, the process waits.

Problems

1. Busy-Waiting: Processes continuously check the lock, wasting CPU time.
2. Starvation: Higher-priority processes dominate resource access.

Alternatives

- Semaphores: Allow processes to wait without using CPU cycles.
- Mutex Locks: Incorporate sleep mechanisms to prevent busy-waiting.

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32. Analyze the critical section problem in concurrent programming.

Critical Section Problem

- Occurs when multiple processes need access to shared resources.
- Must ensure safe access without corrupting data.

Requirements for a Solution

1. Mutual Exclusion: Only one process in the critical section at a time.
2. Progress: Processes not in the critical section should not block others.
3. Bounded Waiting: A process should not wait indefinitely.

Challenges

- Difficult to achieve all three conditions due to system limitations (e.g., starvation).
- Solutions include using synchronization tools like semaphores and locks.

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Sleeping Barber Problem

- Scenario: A barber sleeps if no customers are present. If the shop is full, new customers leave.

Deadlock and Starvation

- Deadlock: Customers and barber wait indefinitely for each other.
- Starvation: Lower-priority customers may never get served.

Solutions

- Use semaphores for managing waiting chairs and barber availability.
- Implement FIFO queues to ensure fairness among customers.

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34. Evaluate the challenges and strategies in multiprocessor scheduling.

Challenges

1. Load Balancing: Evenly distributing tasks across processors.
2. Processor Affinity: Processes prefer to execute on the same CPU for cache performance.
3. Contention: Multiple processes competing for shared resources.

Strategies

- Load Balancing Algorithms: Distribute tasks dynamically or statically.
- Affinity Scheduling: Assign processes to specific CPUs for efficiency.
- Contention Management: Limit simultaneous access to shared resources.

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35. Compare Dekker's and Peterson's solutions for process synchronization.

Dekker's Solution

- Uses flags and turn variables for mutual exclusion.
- Limitations: Complex and unsuitable for modern multiprocessor systems.

Peterson's Solution

- Simpler, uses flags and a turn variable.

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Dekker's Solution

- Uses flags and turn variables for mutual exclusion.
- Limitations: Complex and unsuitable for modern multiprocessor systems.

Peterson's Solution

- Simpler, uses flags and a turn variable.
- Ensures mutual exclusion, progress, and bounded waiting for two processes.
- Limitations: Ineffective for more than two processes or in modern systems.

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36. Compare message passing and shared memory as IPC models in distributed systems.

Message Passing

- Processes communicate via messages.
- Advantages: Secure, suitable for distributed systems.
- Disadvantages: Slower due to communication overhead.

Shared Memory

- Processes access a common memory space.
- Advantages: Faster due to direct memory access.
- Disadvantages: Requires synchronization mechanisms to prevent data inconsistency.
